

1. Project Overview & Core Concept

This document contains the exact architecture and content requirements for a premium, interactive 3D portfolio. The application will feature a stylized 3D map of India. The user's scroll action will drive a cinematic camera fly-through across different regions, triggering specific environment changes (lighting/colors) and revealing HTML text overlays containing the developer's resume details.

2. Tech Stack & Library Requirements

- **Core Framework:** React.js
- **3D Rendering:** Three.js & React Three Fiber (@react-three/fiber, @react-three/drei)
- **Animations & Scroll Control:** GSAP (GreenSock) and ScrollTrigger for camera movement and HTML fading.
- **Styling:** Tailwind CSS for modern, responsive UI overlays and glassmorphism.

3. The 3D Scroll Journey: Zones & Content Mapping

The AI must build a scroll-tied timeline. As the user scrolls, the environment and UI must transition exactly as mapped below:

Zone / Scroll Depth	Environment & Lighting Theme	3D Landmark Placeholders	HTML Overlay Content (Resume Data)
Zone 1: North (Start)	Sky-blue fog with white snow effects. Cool, crisp lighting.	Stylized Mountain Peaks	Hero Section: Adarsh Pathak Full Stack & Django Backend Developer.

Zone / Scroll Depth	Environment & Lighting Theme	3D Landmark Placeholders	HTML Overlay Content (Resume Data)
Zone 2: J&K to Delhi	Transition to lush, vibrant green lighting and environment.	India Gate	Education: MCA @ Lovely Professional University (TGPA: 8.28) B.Sc & B.Ed @ Shri Krishna University.
Zone 3: Uttar Pradesh (Core Hub)	Warm, golden-hour orange/yellow glow. "Majestic Kings of UP" vibe.	Purana Qila, Ram Mandir, Maha Kumbh tents.	Experience & Skills: Django Backend Dev @ TECHQRT Python Web Trainee @ Arudan Skills: Python, Django, DRF, C++, Java, JS, MySQL.
Zone 4: East India	Twilight blue/purple aesthetic.	Howrah Bridge	Projects: MultiShop E-Commerce (Django, MySQL) Hospital Management System (Python).

4. UI/UX Overlay Specifications

- **Navigation:** Create a fixed, floating Glassmorphism navbar at the top with links: Home, Education, Experience, Projects, Skills.
- **Typography:** Use a premium sans-serif font (e.g., Poppins or Inter). Ensure high contrast against the 3D canvas so text is always readable.
- **Transitions:** Text blocks must smoothly fade-in slightly from the bottom when the camera reaches the respective zone, and fade-out as the user scrolls past.

5. Agent Execution Directives

1. **Skip Basic Setup:** Assume Vite, React, and Tailwind are already configured. Jump straight into building the `Canvas` component and the GSAP scroll logic.
2. **Use Placeholders:** Do not attempt to generate complex GLTF models. Use simple Three.js geometric meshes (Boxes, Cylinders) for the landmarks. Structure the code so I can easily swap `` with `` later.
3. **Performance:** Optimize the canvas. Avoid heavy real-time shadows if it drops frame rates; prioritize smooth 60fps scrolling.